

Xing Zhou

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RESEARCH INTERESTS

Limnologist, oceanographer, and modeler with research interests in AI, model development, and applications in ecosystem dynamics and biogeochemical cycling.

EDUCATION

- **Ph.D. in Atmospheric Science**
Michigan Technological University 2017–2023
Dissertation title: *Using Hydrodynamic and Biophysical Models to Study the Mechanisms Mediating Algal Blooms in the Great Lakes*
Advisor: Dr. Pengfei Xue
- **Bachelor of Atmospheric Science**
Lanzhou University 2013-2017

ACADEMIC EMPLOYMENTS

- **Postdoctoral Fellow**
Georgia Institute of Technology 2023–Present
Advisor: Dr. Annalisa Bracco
- **Graduate Research Assistant**
Michigan Technological University 2017-2023

GRANTS & FUNDING

- Co-PI – *Modeling the Dispersal and Connectivity of Marine Larvae with GenAI Agents*. Co-funded by Microsoft and two GeorgiaTech research institutes (the Brook Byers Institute for Sustainable Systems and the Institute for Data Engineering and Science). \$50,000 + \$50,000 Microsoft Azure credit (09/2024–07/2025). (*Served as Co-PI in accordance with Georgia Tech policy that generally does not allow postdoctoral researchers to serve as lead PI, but I conceived, initiated, and led the project.*). PI: A. Bracco; Co-PIs: X. Zhou, R. Wu, J. Abernethy.

PUBLICATIONS

Published

- 1) **Zhou, X.**, Novi, L., Hay, M. E., Montoya, J. P., Aliu, A., Realff, M. J., & Bracco, A. (2026). Changing drivers of the Great Atlantic Sargassum Belt from physical forcing to ecological control. *Nature Communications*.
- 2) **Zhou, X.**, Xiao, S., Ramirez, M., & Bracco, A. (2025). Modeling river and urban related microplastic pollution off the southern United States. *npj Emerging Contaminants*, 1(1), 9.
- 3) Xue, P., Huang, C., Zhong, Y., Notaro, M., Kayastha, M. B., **Zhou, X.**, ... & Kemp, E. (2025). Enhancing winter climate simulations of the Great Lakes: insights from a new coupled lake–ice–atmosphere (CLIAv1) system on the importance of integrating 3D hydrodynamics with a regional climate model. *Geoscientific Model Development*, 18(13), 4293-4316.
- 4) Gardner, S. T., Rowe, M. D., Xue, P., **Zhou, X.**, Alsip, P. J., Bunnell, D. B., ... H'öök, T. O. (2024). Climate-influenced phenology of larval fish transport in a large lake. *Limnology and Oceanography Letters*.
- 5) **Zhou, X.**, Lopera, L., Roa-Varón, A., Bracco, A. (2024). Modeling the larval dispersal and connectivity of Red Snapper (*Lutjanus campechanus*) in the Northern Gulf of Mexico. *Progress in Oceanography*, 224, 103265.

- 6) **Zhou, X.**, Chaffin, J. D., Bratton, J. F., Verhamme, E. M., Xue, P. (2023). Forecasting microcystin concentrations in Lake Erie using a Eulerian tracer model. *Journal of Great Lakes Research*, 49(5), 1029-1044.
- 7) **Zhou, X.**, Rowe, M.D., Liu, Q., Xue, P. (2023). Comparison of Eulerian and Lagrangian transport models for harmful algal bloom forecast in Lake Erie. *Environmental Modelling & Software*, 162, 105641.
- 8) **Zhou, X.**, Auer, M. T., Xue, P. (2021). Open lake phosphorus forcing of *Cladophora* growth: Modeling the dual challenge in great lakes trophic state management. *Water*, 13(19), 2680.
- 9) Chaffin, J. D., Bratton, J. F., Verhamme, E. M., Bair, H. B., Beecher, A. A., Binding, C. E., ... **Zhou, X.** (2021). The Lake Erie HABs Grab: A binational collaboration to characterize the western basin cyanobacterial harmful algal blooms at an unprecedented high-resolution spatial scale. *Harmful Algae*, 108, 102080.
- 10) Xue, P., Schwab, D. J., **Zhou, X.**, Huang, C., Kibler, R., Ye, X. (2018). A hybrid Lagrangian-Eulerian particle model for ecosystem simulation. *Journal of Marine Science and Engineering*, 6(4), 109.

In revision/ Under review

- 1) **Zhou, X.**, Xue, P., Rowe, M. D., Alsip P. J., ... Gardner S. T. Modeling Study of Phytoplankton Responses in Lake Michigan to a Changing Climate. *Limnology and Oceanography*. (in revision)
- 2) **Zhou, X.**, Bracco, A., Ito, T., Reinhard, C. T. Numerical assessment of hypothetical large-scale river-based alkalinity modification scenarios in the northern Gulf of Mexico. *Earth's future* (in revision). Preprint available at <https://cdrxiv.org/preprint/408>
- 3) **Zhou, X.**, Wang, G., Wu, R., & Bracco, A. SWARM: An open-source generative AI agent-based framework for modeling marine larval dispersal and settlement. (submitted to *Nature Machine Intelligence*).

In preparation

- 1) Radtke, H., Rooze, J., Xue, P., **Zhou, X.**, ..., Neumann, T. roboelf v1.0: A GPU-accelerated offline marine biogeochemistry modeling framework – Part 1: General model description. (In preparation for *Geoscientific Model Development*)

PRESENTATIONS (presenter underlined)

- 1) **Zhou, X.** (2025). Modeling Approaches for Sustainable Lake Ecosystems: From Climate Change Impacts to AI-Augmented Frameworks. *Uppsala University Seminar, Sweden*. March 11. (Invited talk)
- 2) **Zhou, X.**, Bracco, A., Ito, T., & Reinhard, C. T. (2025). Numerical assessment of hypothetical large-scale river-based alkalinity modification scenarios in the Northern Gulf Coast. *AGU Fall Meeting 2025*, New Orleans, LA, USA. Dec 15-19. (Poster)
- 3) **Zhou, X.**, Wang, G., Wu, R., Abernethy, J., & Bracco, A. (2025). IchthyopAgent: A generative AI-agent based Lagrangian tool for modeling ichthyoplankton dynamics: Development and application. *Ocean AI Workshop, Halifax Convention Centre, Halifax, Canada*. Nov 4-5. (Talk)
- 4) **Bracco, A.**, Lopera, L., **Zhou, X.**, & Herrera, S. Ecosystem Connectivity in the Northern Gulf: From Corals to Fish. *Benthic Ecology Meeting 2025*, Mobile, AL, April 1–4. (Talk)
- 5) **Zhou, X.**, Wu, R., Wang, G., Bracco, A., Abernethy, J. (2025). Modeling the Dispersal and Connectivity of Marine Larvae with GenAI Agents. *ASLO 2025 Aquatic Science Meeting*, Charlotte, NC, Mar 26-31. (Talk)
- 6) **Ramirez, M.**, **Zhou, X.**, Xiao, S., Bracco, A. (2025). Investigating the Fate of Microplastic Pollution in the Northern Gulf of Mexico and its impact on Marine Habitats. *ASLO 2025 Aquatic Science Meeting*, Charlotte, NC, Mar 26-31. (Poster)
- 7) **Zhou, X.**, Bracco, A., Ito, T., Reinhard, C. (2024) The Impact of Submesoscale Processes on Carbon Dynamics in the Gulf of Mexico during Large-Scale Alkalinity Modification. *AGU fall meeting*, Washington, D.C, Dec 09-13, 2024. (Poster)

- 8) **Zhou, X.**, Xue, P., Rowe, M. D., Alsip P. J., ... Gardner S. T. (2024). Impacts of Climate Change on Phytoplankton Dynamics in Lake Michigan: a Biophysical Modeling Study. *ASLO*, Madison, WI, Jun 2-7, 2024. (Talk)
- 9) **Zhou, X.**, Bracco, A., Lopera, L., & Roa-Varón, A. (2024). Red Snapper (*Lutjanus Campechanus*) Larval Dispersal and Connectivity in the Northern Gulf of Mexico: A High-Resolution Modeling Study. *Ocean science meeting 2024*, New Orleans, Louisiana, Feb 18-23, 2024, (Talk)
- 10) **Zhou, X.**, Xue, P., Rowe, M. D., Alsip P. J., ... Gardner S. T. (2022). Impact of mixing regime shift on phytoplankton, and invasive mussel in Lake Michigan under a warming climate: a biophysical modeling study. *AGU fall meeting*, Chicago, Illinois, Dec 12-16, 2022. (Talk)
- 11) **Zhou, X.**, Xue, P., Chaffin, J., Bratton, J., & Verhamme, E. (2022). Incorporation of microcystin production improves lake erie cyanobacterial bloom toxin forecasts. *In Joint Aquatic Sciences Meeting 2022*. Grand Rapids, Michigan, May 14-20, 2022. (Talk)
- 12) **Zhou, X.**, Xue, P., Liu, Q., & Rowe, M. D. (2021). Intercomparison of three HAB transport models in short-term forecasting of Lake Erie CHABs event. *In 26th Biennial CERF Conference*, Online, 1–4 and 8–11 November 2021. (Talk)
- 13) **Zhou, X.**, Xue, P., & Auer, M. T. (2020). Offshore P-forcing of *Cladophora* growth in the Lake Michigan nearshore: a one-dimensional modeling approach. *International Association for Great Lakes Research*, Online, June 9–11, 2020. (Poster)
- 14) Xue, P., Ye, X., **Zhou, X.**, & Huang, C. (2018). A hybrid Lagrangian-Eulerian particle model for ecosystem simulation in Sandusky Bay. *Estuaries and Coastal Modeling*, Seattle, Washington, June 25-28, 2018. (Talk)
- 15) Kibler, R., Huang, C., **Zhou, X.**, & Xue, P. (2018). Using a property-carrying particle model for ecosystem simulation: A case study of Sandusky Bay. *International Association for Great Lakes Research*, Toronto, Canada. June 18-22, 2018. (Talk)

TEACHING & MENTORING

Teaching

- Teaching Assistant & Co-Lecturer, *Climate Models (EAS 8803)*, Georgia Tech, 2024
- Co-Lecturer, *Introduction to Oceanography (EAS 4300)*, Georgia Tech, 2025; Guest Lecture 2024

Mentoring

- Gabe Lobdell (Undergraduate, University of Georgia, Summer REU, 2025)
- Mireya Ramirez (Undergraduate, Georgia Tech, 2024, PURA Award)
- Aya Kanawati (Undergraduate, Georgia Tech, Spring 2024)

HONORS AND AWARDS

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| ▪ Judith Curry Award, Georgia Institute of Technology | 2024 |
| ▪ Dean's award for outstanding scholarship, Michigan Technological University | 2023 |
| ▪ GLRC Student Research Grant, Michigan Technological University | 2020 |
| ▪ Visiting student in Texas Tech University awarded by China Scholarship Council | 2015 |

PROFESSIONAL SERVICE

- **Editorial Board Member:** *npj Emerging Contaminants*
- **Proposal Reviewer:** NASA Research Initiation Awards (RIA24)
- **Journal Reviewer (selected):** *Progress in Oceanography, Communications Earth & Environment, Journal of Hydrology, Environmental Science & Technology, Scientific Reports*
- **Session Convener & Co-chair:** “Leveraging Biophysical Models to Understand and Mitigate Climate Change Impacts on Aquatic Ecosystems (SS20),” ASLO 2025 Aquatic Sciences Meeting

Outreach & Community Engagement

- Co-hosted interview with Freedom Middle School Robotics Program students on marine biology applications (2024)

- Served as Postdoctoral Representative in faculty hiring processes, School of Earth and Atmospheric Sciences, Georgia Institute of Technology (2024)

MEDIA RECOGNITION

- NCCOS news: Gulf of Mexico Red Snapper Larval Model Suggests Managing Fishery as Two Stocks (<https://coastalscience.noaa.gov/news/gulf-of-mexico-red-snapper-larval-model-suggests-managing-fishery-as-two-stocks/>)
- CMCC news: Research reveals how microplastics threaten Gulf of Mexico marine life (<https://www.cmcc.it/article/research-reveals-how-microplastics-threaten-gulf-of-mexico-marine-life>)